

Hanmeng (Harmony) Zhan

[illegible]

- Bachelor of Arts Jan 2010 - Aug 2012
Department of Economics, Faculty of Arts
Thesis: *Second-Price Auction with Resale*
Supervisor: Philip Curry

Xiamen University, Xiamen, Fujian, China

- Bachelor of Economics Sep 2008 - Jun 2014
Department of Statistics, Faculty of Economics
Thesis: *Multi-Player Multi-State Quantum Games*
Supervisor: Zhengming Qian

PUBLICATIONS **Book**

1. C. Godsil, H. Zhan, *Discrete Quantum Walks on Graphs and Digraphs*. Cambridge University Press (2023).

Peer-Reviewed Papers

2. H. Zhan. ϵ -uniform mixing in discrete quantum walks. *Journal of Algebraic Combinatorics* (2025), 61, 46. doi.org/10.1007/s10801-025-01431-5
3. H. Kumar, B. Mohar, S. Pragada, H. Zhan. *On the second largest adjacency eigenvalue of trees with given diameter*. *Linear and Multilinear Algebra* (2025), 73(13), 2944–2972. <https://doi.org/10.1080/03081087.2025.2484265>
4. H. Zhan. *Simple quantum coins enable pretty good state transfer on every hypercube*. *Philosophical Transactions A: Mathematical, Physical and Engineering Sciences* 383 (2025), no. 2306, 20240418. Theme Issue: Numerical Analysis, Spectral Graph Theory, Orthogonal Polynomials, and Quantum Algorithms. doi.org/10.1098/rsta.2024.0418
5. A. Mohan, H. Zhan. *Discrete quantum walks with marked vertices and their average vertex mixing matrix*. *Linear Algebra and its Applications* 727 (2025), 336–367. doi.org/10.1016/j.laa.2025.08.013
6. H. Kumar, B. Mohar, S. Pragada, H. Zhan. *Subdivision and graph eigenvalues*. *Linear Algebra and its Applications* (2025). doi.org/10.1016/j.laa.2025.01.044
7. Q. Chen, C. Godsil, M. Sobchuk, H. Zhan. *Hamiltonians of bipartite walks*. *Electronic Journal of Combinatorics* (2024) 31(4): P4.10.
8. A. Chan, H. Zhan, *Pretty good state transfer in discrete-time quantum walks*. *Journal of Physics A: Mathematical and Theoretical* (2023). doi:10.1088/1751-8121/acc4f5
9. H. Zhan, *The average search probabilities of discrete-time quantum walks*. *Quantum Information Processing* (2022) 21(9): pp. 336. doi:10.1007/s11128-022-03681-9
10. A. Chan, G. Coutinho, W. Drazen, O. Eisenberg, C. Godsil, M. Kempton, G. Lippner, C. Tamon, H. Zhan, *Fundamentals of fractional revival in graphs*. *Linear Algebra and its Applications* (2022). doi:10.1016/J.LAA.2022.09.010
11. H. Zhan, *Factoring discrete quantum walks on distance regular graphs into continuous quantum walks*. *Linear Algebra and its Applications* (2022), 648: pp. 88–103. doi:10.1016/J.LAA.2022.04.017

12. A. Chan, B. Johnson, M. Liu, M. Schmidt, Z. Yin, H. Zhan, *Laplacian pretty good fractional revival*. Discrete Mathematics (2022), 345(10), 112971. doi:10.1016/J.DISC.2022.112971
13. A. Chan, B. Johnson, M. Liu, M. Schmidt, Z. Yin, H. Zhan, *Laplacian fractional revival on graphs*. Electronic Journal of Combinatorics (2021) 28(3): P3.22.
14. A. Chan, G. Coutinho, C. Tamon, L. Vinet, H. Zhan, *Fractional revival and association schemes*. Discrete Mathematics (2020) 343(11), 112018. doi:10.1016/j.disc.2020.112018
15. L. Vinet, H. Zhan, *Perfect state transfer on weighted graphs of the Johnson scheme*. Letter in Mathematical Physics (2020). doi.org/10.1007/s11005-020-01298-6
16. H. Zhan, *Quantum walks on embeddings*. Journal of Algebraic Combinatorics (2020). doi.org/10.1007/s10801-020-00958-z
17. H. Zhan, *An infinite family of circulant graphs with perfect state transfer in discrete quantum walks*. Quantum Information Processing (2019) 18(12): pp. 369.
18. G. Coutinho, L. Vinet, H. Zhan, A. Zhedanov. *Perfect state transfer in a spin chain without mirror symmetry*. Journal of Physics A: Mathematical and Theoretical (2019) 52(45), pp. 455302.
19. C. Godsil, H. Zhan, *Discrete-time quantum walks and graph structures*. Journal of Combinatorial Theory, Series A (2019), pp. 181–212.
20. A. Chan, G. Coutinho, C. Tamon, L. Vinet, H. Zhan, *Quantum fractional revival on graphs*. Discrete Applied Mathematics (2019) 269, pp. 86-98.
21. G. Coutinho, C. Godsil, K. Guo, H. Zhan, *A new perspective on the average mixing matrix*. Electronic Journal of Combinatorics (2018) 25(4): P4.14.
22. C. Godsil, H. Zhan, *Uniform mixing on Cayley graphs*. Electronic Journal of Combinatorics (2017) 24(3): P3.20.
23. G. Coutinho, C. Godsil, M. Shirazi, H. Zhan, *Equiangular lines and covers of the complete graph*. Linear Algebra and its Applications (2016) 488: pp. 264-283.
24. R. Alvir, S. Dever, B. Lovitz, J. Myer, C. Tamon, Y. Xu, H. Zhan. *Perfect state transfer in Laplacian quantum walk*. Journal of Algebraic Combinatorics (2016) 43(4): pp. 801-826.

Submitted preprints

25. Y. Xu, H. Zhan. *Subspace state transfer in coined quantum walks*. arXiv:2511.06750v1 (2025).

MENTORING EXPERIENCE

Graduate Research Programs

- WPI Graduate Independent Studies Project Jan - Present, 2024
Supervising graduate research projects on discrete quantum walks.

Undergraduate Research Programs

- WPI Major Qualifying Projects May 2024 - Present
Supervising senior capstone projects on discrete quantum walks.

- WPI STAR Summer Research Training Program May - Aug, 2024
Supervised an undergraduate research project on continuous quantum walks.
- Fields Undergraduate Summer Research Program Jul - Aug, 2020
Co-supervised an undergraduate research project with Ada Chan on Laplacian fractional revival at the Fields Institute. This results in two papers [6] and [7]. One student was awarded the Book Prize by the Fields Institute.
- University of Waterloo May - Aug, 2018
Mentored one undergraduate student under the supervision of Chris Godsil. The student extended my results in the paper [10], and developed the theory of a new quantum walk called the vertex-face walk. This leads to the joint work [19].

PRESENTATIONS **Invited Talks**

1. *Laziness of quantum walks*. In: Canadian Mathematical Society Winter Meeting, Toronto, Ontario, December 5 - 8, 2025.
2. *Quantum state transfer in association schemes*. In: Combinatorics around q-Onsager algebra, Kranjska Gora, Slovenia, July 23 - 27, 2025.
3. *New connections between quantum walks and graph spectra*. In: CMS Summer Meeting, Université Laval, Québec, QC, Canada, June 6 - 9, 2025.
4. *Average mixing matrices of quantum walks*. In: Canadian Discrete and Algorithmic Mathematics Conference, University of Ottawa, Ottawa, ON, Canada, May 20 - 23, 2025.
5. *Discrete quantum walks in association schemes*. In: Canadian Discrete and Algorithmic Mathematics Conference, University of Ottawa, Ottawa, ON, Canada, May 20 - 23, 2025.
6. *Quantum state transfer with fair and unfair coins*. In: American Mathematical Society Sectional Meeting, University of Connecticut, Hartford, CT United States, April 5 - 6, 2025.
7. *Coined quantum walks and weighted adjacency matrices*. In: Joint Mathematics Meetings, Seattle Convention Center, Seattle, WA, United States, January 8 - 11, 2025.
8. *Limiting behavior of coined quantum walks with marked vertices*. In: Canadian Mathematical Society Winter Meeting, Kwantlen Polytechnic University, Vancouver, BC, Canada, November 29 - December 2, 2024.
9. *Recent progress in coined quantum walks*. In: Movement and Symmetry in Graphs, Banff International Research Station for Mathematical Innovation and Discovery, Banff, AB, Canada, November 24 - 29, 2024.
10. *Walking on graphs with a quantum coin*. In: Math Colloquium, University of Manitoba, Winnipeg, MB, Canada, November 21, 2024.
11. *Laplacian fractional revival on graphs*. In: Math Colloquium, University of Hartford, Hartford, CT, United States, October 4, 2024.
12. *Epsilon-uniform mixing on strongly regular graphs via coined quantum walks*. In: 05C50 Online, University of Manitoba, Winnipeg, MB, Canada, September 13, 2024.

13. *Generating quantum uniform mixing in association schemes.* In: CMS Summer Meeting, University of Saskatchewan, Saskatoon, SK, Canada, May 31 - June 3, 2024.
14. *Discrete quantum walks in schemes.* In: AMS Sectional Meeting, University of Wisconsin-Milwaukee, Milwaukee, WI, United States, April 20 - 21, 2024.
15. *Quantum walks: from continuous to discrete.* In: Godsil75, University of Waterloo, Waterloo, ON, Canada, March 15 - 17, 2024.
16. *Strongly cospectral vertices and their phantom mates.* In: Joint Mathematics Meetings, Moscone Center, San Francisco, CA, United States, January 3 - 6, 2024.
17. *Quantum search: an averaging perspective.* In: Joint Mathematics Meetings, Moscone Center, San Francisco, CA, United States, January 3 - 6, 2024.
18. *Spectra of line digraphs and their applications.* In: CMS Winter Meeting, Montréal, QC, Canada, December 1 - 4, 2023.
19. *How graph spectra determine the behavior of discrete quantum walks.* In: 10th International Workshop of Quantum Simulation and Quantum Walks, Tsukuba International Congress Center, Tsukuba, Japan, November 10 - 12, 2023.
20. *On the second largest eigenvalue of a tree.* In: On-line Combinatorics Seminar, University of Wisconsin-Madison, Madison, WI, October 23, 2023.
21. *Some recent results in quantum walks.* In: Algebraic structures and special functions in theoretical physics, Ghent University, Ghent, Belgium, June 26 - 30, 2023.
22. *Discrete quantum walks in association schemes.* In: 10th Slovenian Conference on Graph Theory, Kranjska Gora, Slovenia, June 18 - 24, 2023.
23. *Some open problems in discrete quantum walks.* In: Canadian Discrete and Algorithmic Mathematics Conference, University of Manitoba and University of Winnipeg, Winnipeg, MB, Canada, June 5 - 8, 2023.
24. *Discrete quantum walks on distance regular graphs.* In: Canadian Discrete and Algorithmic Mathematics Conference, University of Manitoba and University of Winnipeg, Winnipeg, MB, Canada, June 5 - 8, 2023.
25. *The second largest eigenvalue of a tree.* In: CMS Winter Meeting, Chelsea Hotel, Toronto, ON, Canada, December 2 - 5, 2022.
26. *The effect of marking vertices in discrete quantum walks.* In: Graph Theory, Algebraic Combinatorics and Mathematical Physics, Centre de Recherches Mathématiques, July 25 - August 19, 2022.
27. *The average search probability in a quantum walk with an oracle.* In: Algebraic Graph Theory Seminar, University of Waterloo, Waterloo, ON, Canada, August 2, 2021.
28. *Arc-reversal quantum walks.* In: Discrete Math Seminar, Simon Fraser University, Vancouver, BC, Canada, February 24, 2021.
29. *DRACKNs and their applications in quantum information.* In: Codes and Expansions, United States, September 8, 2020.
30. *Factoring discrete quantum walks into continuous quantum walks.* In: Algebraic Graph Theory Seminar, University of Waterloo, Waterloo, ON, Canada, August 3, 2020.

31. *Quantum fractional revival*. In: Discrete Math Seminar, University of Delaware, Newark, DE, United States, April 23, 2020.
32. *New advances in quantum walks*. In: AMS Joint Mathematics Meetings, Colorado Convention Center, Denver, CO, United States, January 15 - 18, 2020.
33. *State transfer via orthogonal polynomials*. In: AMS Sectional Meeting, University of Wisconsin-Madison, Madison, WI, United States, September 14 - 15, 2019.
34. *Quantum state transfer in the algebra of the Johnson scheme*. In: CMS Summer Meeting, University of Regina, Regina, SK, Canada, June 7 - 10, 2019.
35. *Some elegant results in algebraic graph theory*. In: Canadian Discrete and Algorithmic Mathematics Conference, Simon Fraser University, Vancouver, BC, Canada, May 28 - 31, 2019.
36. *Quantum walks, orthogonal polynomials, and spectral graph theory*. In: Quantum Walks and Information Tasks, Banff International Research Station for Mathematical Innovation and Discovery, Banff, AB, Canada, April 21 - 26, 2019.
37. *Generating entanglement using quantum walks*. In: David A. Walsh Seminar Series, Clarkson University, Potsdam, NY, United States, February 8, 2019.
38. *Some open problems in discrete quantum walks*. In: Algebraic Graph Theory and Quantum Walks, University of Waterloo, Waterloo, ON, Canada, April 23 - 27, 2018.
39. *Recent progress in discrete quantum walks*. In: AMS Sectional Meeting, Northeastern University, Boston, MA, United States, April 21 - 22, 2018, 2018.
40. *Graph covers and equiangular frames*. In: AMS Sectional Meeting, Ohio State University, Columbus, OH, United States, March 16 - 18, 2018.
41. *From covers to tight frames*. In: AMS Sectional Meeting, College of Charleston, Charleston, SC, United States, March 10 - 12, 2017.
42. *Spectra of discrete quantum walks*. In: CMS Summer Meeting, University of Alberta, Edmonton, AB, Canada, June 24 - 27, 2016.
43. *Lines and covers of complete graphs 2*. In: Systems of Lines: Applications of Algebraic Combinatorics, Worcester Polytechnic Institute, Worcester, MA, United States, August 10 - 14, 2015.
44. *Some open problems in uniform mixing*. In: Summer Research Program, Clarkson University, Potsdam, NY, United States, July 20, 2015.

Mini-Course

45. *Introduction to discrete quantum walks*. In: CMS Winter Meeting, Chelsea Hotel, Toronto, ON, Canada, December 2 - 5, 2022.

Contributed Talks

46. *Average mixing in discrete quantum walks with marked vertices*. In: AMS Fall Meeting, Albany, MA, USA, October 19 - 20, 2024.
47. *ϵ -uniform mixing in discrete quantum walks*. In: CMS Winter Meeting, Montréal, QC, Canada, December 1 - 4, 2023.

48. *Discrete quantum walk search on graphs*. In: Coast Combinatorics Conference 2023, SFU Harbour Centre, Vancouver, BC, Canada, March 4 - 5, 2023.
49. *How far can the quantum walker go*. In: 9th International Conference on Quantum Simulation and Quantum Walks, Centre International de Rencontres Mathématiques, Marseille, Bouches-du-Rhone, France, January 20 - 24, 2020.
50. *Discrete quantum walks on Cayley graphs*. In: CMS Winter Meeting, Chelsea Hotel, Toronto, Toronto, ON, Canada, December 6 - 9, 2019.
51. *The vertex-face walk*. In: Finite Geometry and Extremal Combinatorics, University of Delaware, Newark, DE, United States, August 21 - 24, 2019.
52. *Combinatorial aspects of quantum walks*. In: Prairie Discrete Math Workshop, Brandon University, Brandon, MB, Canada, June 12 - 15, 2018.
53. *Discrete-time quantum walks and graph embeddings*. In: CMS Winter Meeting, University of Waterloo, Waterloo, ON, Canada, December 8- 11, 2017.
54. *Quantum walks and mixing*. In: Algebraic and Extremal Graph Theory, University of Delaware, Newark, DE, United States, August 7 - 10, 2017.
55. *Discrete-time quantum walks and graph structures*. In: Canadian Discrete and Algorithmic Mathematics Conference, Ryerson University, Toronto, ON, Canada, June 12 - 15, 2017.
56. *Uniform mixing in quantum walks*. In: 22nd Ontario Combinatorics Workshop, York University, Toronto, ON, Canada, May 16 - 17, 2014.

TEACHING
EXPERIENCE

Worcester Polytechnic Institute, Worcester, MA, USA

- Instructor Fall A-term 2025
CS 3133: Foundations of Computer Science (class size: 50)
- Instructor Spring D-term 2024, 2025
CS 4123: Theory of Computation (class size: 40)
- Instructor Spring 2024, 2025
CS 503: Foundations of Computer Science (class size: 25)

Simon Fraser University, Vancouver, BC, Canada

- Instructor Spring 2023
MACM 201: Discrete Mathematics II (class size: 180)
- Instructor Summer 2022
MATH 240: Algebra I: Linear Algebra (class size: 90)

Online Graduate Courses, International

- Instructor Winter 2021
Combinatorics and Quantum Walks (class size: 20; recordings available)

York University, Toronto, ON, Canada

- Instructor Winter 2020, Fall 2020
MATH 1014: Applied Calculus II (class size: 180 - 280)

University of Waterloo, Waterloo, ON, Canada

- Instructor Winter 2018
MATH 135: Algebra for Honors Mathematics (class size: 60)
- Guest Lecturer Winter 2017
CO 444/644: Algebraic Graph Theory (class size: 20)

FUNDING,
AWARDS AND
DISTINCTIONS

Federal Research Grants

- NSF CRII: FET: Quantum Advantages through Discrete Quantum Walks 2024 - 2026
Award number: 2348399
Amount: \$174420.00

Funding for Conferences Organized

- Algebraic Graph Theory and Quantum Information 2021

Type and Source	Amount in CAD	Purpose
Fields Institute	\$14250	travel funding for non-US invited speakers, at least \$5000 towards ECRs, females, postdocs, students
NSF, via Fields Institute	\$10000	travel funding for US invited speakers, at least 2/3 towards ECRs, females, postdocs, students
York University, via Office of Vice-President Research and Innovation	\$2000	registration fees for students and postdocs

Table 1: Funding for Algebraic Graph Theory and Quantum Information

Awards and Distinctions from University of Waterloo

- *University Finalist for the Governor General’s Gold Medal* 2019
Purpose: a medal awarded to nominees for the Governor General’s Gold Medal
- *Inaugural Mathematics Doctoral Prize, First Prize* 2019
Purpose: a prize awarded to recognize the achievement of graduating doctoral students in the Faculty of Mathematics
Amount: \$1500
- *Outstanding Achievement in Graduate Studies* 2015
Purpose: an honor awarded to three University of Waterloo Master’s students for their outstanding achievement in graduate studies
- *Cotton Family Women in Mathematics Graduate Scholarship* 2014, 2016, 2017
Purpose: a scholarship awarded to a full-time female graduate student on the basis of academic excellence in their studies and research
Amount: \$9000
- *Robin K. Banks Scholarship* 2011 - 2012
Purpose: a scholarship awarded to a full-time student in the Faculty of Arts who have achieved the highest overall average at the end of Year Three

Amount: \$750

- *Faculty of Arts Upper-Year Scholarship* 2011 - 2012
Purpose: a scholarship awarded to outstanding full-time and part-time students in the Faculty of Arts on the basis of overall average
Amount: \$500
- *Dean's Honours List* 2011 - 2012

SERVICE

Conference, Workshop and Seminar Organizer

- Algebraic Graph Theory and Applications, BIRS October 12 - 17, 2025
Co-organizers: Ada Chan, Gabriel Coutinho, Chris Godsil
- Spectral Graph Theory, Orthogonal Polynomials, and Quantum Computing, Third Joint SIAM/CAIMS annual meeting July 28 - August 1, 2025
Co-organizer: Anastasiia Minenkova
- Quantum Information on Graphs, CANADAM May 20 - 23, 2025
Co-organizers: Thomás Jung Spier, Xiaohong Zhan
- Algebraic Graph Theory, CMS Winter Meeting Nov 29 - Dec 2, 2024
Co-organizers: Sooyeong Kim, Hermie Monderde, Bidy Razafimahatratra, Thomas Jung Spier, Xiaohong Zhang
- Quantum Mathematics for Computation, AMS Fall Sectional Meeting Oct 19 - 20, 2024
Co-organizer: Christino Tamon
- QIT Thinking Seminar, Worcester Polytechnic Institute 2024
Co-organizer: William Martin
- Quantum Information on Graphs, Women in Combinatorics Virtual Conference Jul 15 - 16, 2024
Co-organizer: Xiaohong Zhang
- Algebraic Graph Theory for Walking on Graphs, CMS Winter Meeting 2023
Co-organizers: Sooyeong Kim, Hermie Monderde, Christopher Van Bommel, Xiaohong Zhang
- Algebraic Graph Theory and Quantum Information, Fields Institute 2021
Co-organizers: Ada Chan, Gabriel Coutinho, Krystal Guo, Christino Tamon, Luc Vinet
- Quantum Information on Graphs, CMS Winter Meeting Dec 6 - 9, 2019
Co-organizers: Ada Chan, Christino Tamon
- Discrete Mathematics Seminar, York University Fall 2019
Co-organizer: Justin M. Troyka

Seminar Chair

- Algebraic Graph Theory Seminar, University of Waterloo Spring 2016, Fall 2017

Journal Guest Editor

- Special Issue "Numerical Analysis, Spectral Graph Theory, Orthogonal Polynomials, and Quantum Algorithms" of Philosophical Transactions A 2024

Editors: Anastasiia Minenkova, Gamal Mograby

Journal Reviewer

- Algorithmica 2024
- Discrete Mathematics 2021 - 2024
- Electronic Journal of Combinatorics 2017 - 2021, 2024
- Linear Algebra and Its Applications 2016, 2019 - 2024
- Quantum Information Processing 2022 - 2023
- Linear and Multilinear Algebra 2019 - 2020, 2023
- Algebraic Combinatorics 2022
- Journal of Combinatorial Theory, Series A 2021
- Communications in Algebra 2019
- Journal of Physics A: Mathematical and Theoretical 2019

Conference Reviewer

- Sampling Theory and Applications 2019
- International Colloquium on Automata, Languages and Programming 2018